A disk theorem for the h_1-window, h_k-eigenfunction problem

1 Purpose of this note

The goal of this note is to record, in a fully explicit and deliberately detailed way, a rigidity argument for the localization problem in which the window is h_1 and the eigenfunction is the k-th Hermite function h_k for $k \geq 1$.

The argument begins from the original localization identity, passes to the Bargmann–Fock model, derives a compactly supported Poisson problem by a Paley–Wiener divisibility argument, and then shows that, *provided the compactly supported potential has the boundary regularity used below and the anti-holomorphic quantity constructed near the end vanishes to order at least $k-1$ at the origin*, a simply connected domain must be a disk centered at the origin.

So the logical structure is:

(1) rewrite the localization eigenvalue equation in Bargmann form;

(2) derive from it a compactly supported Poisson problem;

(3) invoke an explicit divisibility lemma for compactly supported distributions;

(4) identify and remove the higher-order singular term explicitly;

(5) identify a primitive for the smooth bulk source;

(6) construct an anti-holomorphic quantity;

(7) use the compact-support consequence to show that a related holomorphic quantity is real on the boundary;

(8) conclude that it is constant;

(9) deduce that the boundary lies on a radial level set;

(10) use simple connectivity to conclude that the domain is a disk.

2 From localization to a compactly supported Poisson problem

We assume throughout that $U \subset \mathbb{C}$ is a bounded simply connected domain of class C^2 and that $0 \in U$.

Let $\mathcal{L}_U^{h_1}$ denote the time-frequency localization operator associated with the domain U and window h_1. We assume that h_k is an eigenfunction:

$$\mathcal{L}_U^{h_1} h_k = \lambda h_k$$

for some $\lambda \in \mathbb{R}$.

Under the Bargmann transform, h_k corresponds to the normalized monomial

$$e_k(w) = \sqrt{\frac{\pi^k}{k!}}\, w^k.$$

If T_z denotes the Weyl operator on Fock space, then

$$(T_z F)(w) = e^{\pi \bar{z} w - \frac{\pi}{2}|z|^2} F(w - z),$$

and for $F = e_1(w) = \sqrt{\pi} w$, we have

$$T_z e_1(w) = \sqrt{\pi}\, e^{\pi \bar{z} w - \frac{\pi}{2}|z|^2} (w - z).$$

One computes the inner product

$$\langle e_k, T_z e_1 \rangle = \sqrt{\frac{\pi^{k+1}}{k!}} e^{-\frac{\pi}{2}|z|^2} \int_{\mathbb{C}} \zeta^k (\bar{\zeta} - \bar{z}) e^{\pi z \bar{\zeta}} e^{-\pi|\zeta|^2}\, dA(\zeta).$$

Using the reproducing property and its derivative,

$$\int_{\mathbb{C}} \zeta^k e^{\pi z \bar{\zeta}} e^{-\pi|\zeta|^2}\, dA(\zeta) = z^k, \qquad \int_{\mathbb{C}} \zeta^k \bar{\zeta} e^{\pi z \bar{\zeta}} e^{-\pi|\zeta|^2}\, dA(\zeta) = \frac{k}{\pi} z^{k-1}.$$

Thus

$$\langle e_k, T_z e_1 \rangle = \sqrt{\frac{\pi^{k+1}}{k!}} z^{k-1} e^{-\frac{\pi}{2}|z|^2} \left(\frac{k}{\pi} - |z|^2 \right).$$

Consequently, the eigenvalue equation is equivalent to

$$\int_U \frac{\pi^{k+1}}{k!} z^{k-1} \left(\frac{k}{\pi} - |z|^2 \right) (w - z) e^{\pi \bar{z} w} e^{-\pi|z|^2}\, dA(z) = \lambda \sqrt{\frac{\pi^k}{k!}} w^k. \tag{1}$$

Simplifying constants and taking complex conjugates with replacement $\bar{w} \to w$, we obtain

$$\int_U \bar{z}^{k-1} (k - \pi|z|^2)(w - \bar{z}) e^{\pi w z} e^{-\pi|z|^2}\, dA(z) = \Lambda w^k, \qquad \Lambda = \frac{\lambda \sqrt{k!}}{\pi \sqrt{\pi^k}}. \tag{2}$$

Lemma 2.1 (Paley–Wiener divisibility for Δ)**.** *Let $\omega \in \mathcal{E}'(\mathbb{R}^2)$ be a compactly supported distribution such that*

$$\widehat{\omega}(w, \pm iw) = 0 \qquad \text{for every } w \in \mathbb{C}.$$

Then there exists a compactly supported distribution $u \in \mathcal{E}'(\mathbb{R}^2)$ such that

$$\Delta u = \omega$$

in the sense of distributions.

Proof. By the Paley–Wiener theorem, $\widehat{\omega}(\xi, \eta)$ is an entire function on $\mathbb{C}^2$ of exponential type. Write

$$\widehat{\omega}(\xi, \eta) = \sum_{n \geq 0} H_n(\xi, \eta),$$

where H_n is the homogeneous polynomial of degree n.

The vanishing assumptions on the characteristic lines imply

$$H_n(w, \pm iw) = 0 \qquad \text{for every } w \in \mathbb{C} \text{ and every } n \geq 0.$$

Hence each H_n vanishes on the two complex lines $\eta = \pm i\xi$, so each H_n is divisible by $\xi^2 + \eta^2$. Thus

$$H_n(\xi, \eta) = (\xi^2 + \eta^2) G_{n-2}(\xi, \eta)$$

for some homogeneous polynomial G_{n-2}, with $G_{n-2} \equiv 0$ for $n < 2$.

Summing over n, we obtain an entire function

$$F(\xi,\eta) := \sum_{n\geq 2} G_{n-2}(\xi,\eta)$$

such that

$$\widehat{\omega}(\xi,\eta) = (\xi^2+\eta^2)F(\xi,\eta).$$

Since $\widehat{\omega}$ has exponential type, so does F. By Paley–Wiener again, there exists a compactly supported distribution $u \in \mathcal{E}'(\mathbb{R}^2)$ whose Fourier transform is $-F$.

We then have

$$\widehat{\Delta u}(\xi,\eta) = -(\xi^2+\eta^2)\widehat{u}(\xi,\eta) = \widehat{\omega}(\xi,\eta),$$

and therefore $\Delta u = \omega$. □

Proposition 2.2. *There exists a compactly supported distributional solution u on $\mathbb{C}$ such that*

$$\Delta u = \bar{w}^k e^{-\pi|w|^2}\mathbf{1}_U - \frac{1}{\pi}\bar{w}^{k-1}(k-\pi|w|^2)e^{-\pi|w|^2}\partial_w \mathbf{1}_U - \frac{\Lambda}{(-1)^k\pi^k}\partial_w^k\delta_0. \tag{3}$$

Proof. Set

$$g(z) := \bar{z}^{k-1}(k-\pi|z|^2)\, e^{-\pi|z|^2}.$$

Consider the compactly supported distribution

$$\omega := \left(-\frac{1}{\pi}\partial_w - \bar{w}\right)(g(w)\mathbf{1}_U) - \frac{\Lambda}{(-1)^k\pi^k}\partial_w^k\delta_0.$$

We show that $\langle \omega, e^{\pi wz}\rangle = 0$ for every $w \in \mathbb{C}$.

Pairing the first two terms against $e^{\pi wz}$ (with z as the distributional variable) and using the definition of distributional derivative gives

$$\left\langle \left(-\frac{1}{\pi}\partial_w - \bar{z}\right)(g(z)\mathbf{1}_U)\,,\, e^{\pi wz}\right\rangle = \int_U g(z)\,(w-\bar{z})\, e^{\pi wz}\, dA(z).$$

For the delta term, using $\langle \partial_w^k\delta_0, \varphi\rangle = (-1)^k\partial_w^k\varphi(0)$,

$$\left\langle -\frac{\Lambda}{(-1)^k\pi^k}\,\partial_w^k\delta_0,\, e^{\pi wz}\right\rangle = \frac{-\Lambda}{(-1)^k\pi^k}\cdot(-1)^k\pi^k w^k = -\Lambda w^k.$$

Therefore

$$\langle \omega, e^{\pi wz}\rangle = \int_U \bar{z}^{k-1}(k-\pi|z|^2)(w-\bar{z})\, e^{\pi wz}\, e^{-\pi|z|^2}\, dA(z) - \Lambda w^k.$$

By (2), the right-hand side vanishes for every $w \in \mathbb{C}$, so

$$\langle \omega, e^{\pi wz}\rangle = 0 \qquad \text{for every } w \in \mathbb{C}.$$

Since

$$-\frac{1}{\pi}\partial_w g(w) - \bar{w}\, g(w) = \bar{w}^k e^{-\pi|w|^2},$$

we may rewrite ω as

$$\omega = \bar{w}^k e^{-\pi|w|^2}\mathbf{1}_U - \frac{1}{\pi}\bar{w}^{k-1}(k-\pi|w|^2)e^{-\pi|w|^2}\partial_w\mathbf{1}_U - \frac{\Lambda}{(-1)^k\pi^k}\partial_w^k\delta_0.$$

Writing $z = x+iy$, this means the Fourier transform of ω, viewed as an entire function on $\mathbb{C}^2$ via Paley–Wiener–Schwartz, vanishes on the characteristic lines $\widehat{\omega}(w, \pm iw) = 0$. Lemma 2.1 therefore yields a compactly supported distribution u on $\mathbb{C}$ with

$$\Delta u = \omega,$$

which is exactly (3). □

3 Conventions and distributional identities

We fix the standard Wirtinger operators

$$\partial_w = \frac{1}{2}(\partial_x - i\partial_y), \qquad \partial_{\bar{w}} = \frac{1}{2}(\partial_x + i\partial_y),$$

so that $\Delta = 4\partial_w\partial_{\bar{w}}$. Recall $\partial_{\bar{w}}(1/w) = \pi\delta_0$ and the fundamental solution $G(w) = \frac{1}{2\pi}\log|w|$.

4 The singular part coming from $\partial_w^k\delta_0$

Lemma 4.1. *Define*

$$u_{\text{sing}}(w) := -\frac{\Lambda}{(-1)^k\pi^k}\partial_w^k G(w). \tag{4}$$

Then $\Delta u_{\text{sing}} = -\frac{\Lambda}{(-1)^k\pi^k}\partial_w^k\delta_0$. *For* $w \neq 0$,

$$u_{\text{sing}}(w) = \frac{\Lambda(k-1)!}{4\pi^{k+1}w^k}. \tag{5}$$

Proof. Direct differentiation of $G(w) = \frac{1}{2\pi}\log|w|$. Since $\partial_w \log|w| = 1/(2w)$, it follows that $\partial_w^k \log|w| = \frac{(-1)^{k-1}(k-1)!}{2w^k}$. Combining the constants gives (5). □

5 Boundary consequences of compact support

Lemma 5.1. *Assume u is the compactly supported solution to* (3). *The function u is harmonic on $\mathbb{C}\setminus(\overline{U}\cup\{0\})$, and in fact*

$$u \equiv 0 \qquad \textit{on the unbounded component of } \mathbb{C}\setminus\overline{U}.$$

Assume in addition that $u|_U \in C^{1,\alpha}(\overline{U})$ for some $\alpha \in (0,1)$. Then on ∂U,

$$u = 0, \qquad \partial_{\bar{w}}u = -\frac{1}{4\pi}\bar{w}^{k-1}(k-\pi|w|^2)e^{-\pi|w|^2}. \tag{6}$$

Proof. On the unbounded connected component of $\mathbb{C}\setminus\overline{U}$, the right-hand side of (3) vanishes, so u is harmonic there. Since u is compactly supported, it vanishes identically in a neighborhood of infinity. By unique continuation for harmonic functions, $u \equiv 0$ on the whole unbounded component of $\mathbb{C}\setminus\overline{U}$.

The trace $u = 0$ on ∂U follows immediately.

We now read off $\partial_{\bar{w}}u$ on ∂U. Since $u = 0$ on ∂U, one has distributionally

$$\partial_{\bar{w}}u = \mathbf{1}_U\,\partial_{\bar{w}}u.$$

Applying $4\partial_w$, we obtain

$$\Delta u = 4\partial_w(\mathbf{1}_U\,\partial_{\bar{w}}u) = \mathbf{1}_U\,\Delta u + 4(\partial_w\mathbf{1}_U)\,\partial_{\bar{w}}u.$$

On the other hand, expanding (3) and noting that

$$-\frac{1}{\pi}\partial_w\Big(\bar{w}^{k-1}(k-\pi|w|^2)e^{-\pi|w|^2}\Big) = \bar{w}^k e^{-\pi|w|^2}$$

holds inside U, we can write Δu as

$$\Delta u = \bar{w}^k e^{-\pi|w|^2}\mathbf{1}_U - \frac{1}{\pi}\bar{w}^{k-1}(k-\pi|w|^2)e^{-\pi|w|^2}\partial_w\mathbf{1}_U - \frac{\Lambda}{(-1)^k\pi^k}\partial_w^k\delta_0.$$

Comparing the coefficients of the boundary distribution $\partial_w\mathbf{1}_U$ in the two expressions for Δu, we conclude that

$$4\,\partial_{\bar{w}}u = -\frac{1}{\pi}\bar{w}^{k-1}(k-\pi|w|^2)e^{-\pi|w|^2} \qquad \text{on } \partial U,$$

which is exactly (6). □

6 Removal of the singular source

Definition 6.1. *Define* $v := u - u_{\text{sing}}$.

Lemma 6.2. *The function* v *satisfies* $\Delta v = \bar{w}^k e^{-\pi|w|^2}$ *in* U. *In particular,* $v \in C^\infty_{\text{loc}}(U)$, *and* v *is smooth near* 0.

Proof. By (3), Lemma 4.1, and the definition $v = u - u_{\text{sing}}$,

$$\Delta v = \Delta u - \Delta u_{\text{sing}} = \left(\bar{w}^k e^{-\pi|w|^2}\mathbf{1}_U - \frac{1}{\pi}\bar{w}^{k-1}(k - \pi|w|^2)e^{-\pi|w|^2}\partial_w \mathbf{1}_U - \frac{\Lambda}{(-1)^k\pi^k}\partial_w^k\delta_0\right) - \left(-\frac{\Lambda}{(-1)^k\pi^k}\partial_w^k\delta_0\right).$$

The two delta terms cancel. Away from the boundary of U, the term involving $\partial_w \mathbf{1}_U$ also vanishes, leaving

$$\Delta v = \bar{w}^k e^{-\pi|w|^2} \qquad \text{in } U.$$

Since the right-hand side is C^∞ in U, elliptic regularity gives $v \in C^\infty_{\text{loc}}(U)$, and in particular v is smooth near 0. □

7 An explicit primitive of the smooth bulk source

Lemma 7.1. *Define*

$$F_0(w, \bar{w}) := -\frac{1}{4\pi}\bar{w}^{k-1}e^{-\pi|w|^2}. \tag{7}$$

Then $4\partial_w F_0 = \bar{w}^k e^{-\pi|w|^2}$.

Proof. Compute $4\partial_w F_0 = 4(-\frac{1}{4\pi})\bar{w}^{k-1}(-\pi\bar{w})e^{-\pi|w|^2} = \bar{w}^k e^{-\pi|w|^2}$. □

8 Construction of the anti-holomorphic quantity

Definition 8.1. *Set* $q(w) := \partial_{\bar{w}} v(w) - F_0(w, \bar{w})$.

Proposition 8.2. *The function* q *is anti-holomorphic in* U.

Proof. By definition, $q = \partial_{\bar{w}} v - F_0$. Apply $4\partial_w$ to both sides:

$$4\partial_w q = 4\partial_w \partial_{\bar{w}} v - 4\partial_w F_0.$$

Using Lemma 6.2, $4\partial_w\partial_{\bar{w}} v = \Delta v = \bar{w}^k e^{-\pi|w|^2}$. Using Lemma 7.1, $4\partial_w F_0 = \bar{w}^k e^{-\pi|w|^2}$. Subtracting, $4\partial_w q = 0$, so q is anti-holomorphic in U.

Since v is smooth in U by Lemma 6.2 and F_0 is smooth everywhere, q is smooth on all of U, including near 0. □

9 Boundary values of q

Lemma 9.1. *On* ∂U,

$$q(w) = \frac{1}{4\pi}\bar{w}^{k-1}e^{-\pi|w|^2}(1 - k + \pi|w|^2). \tag{8}$$

Proof. We use $v = u - u_{\text{sing}}$. Since $u_{\text{sing}}(w) = \Lambda(k-1)!/(4\pi^{k+1}w^k)$ is holomorphic away from 0, and $0 \in U$ so $0 \notin \partial U$, we have

$$\partial_{\bar{w}} u_{\text{sing}} = 0 \qquad \text{on } \partial U.$$

Therefore $\partial_{\bar{w}} v = \partial_{\bar{w}} u$ on ∂U.

Using the boundary condition from Lemma 5.1,

$$\partial_{\bar{w}} u = -\frac{1}{4\pi}\bar{w}^{k-1}(k-\pi|w|^2)e^{-\pi|w|^2} \qquad \text{on } \partial U.$$

By definition of q and the formula for F_0,

$$\begin{aligned} q(w) &= \partial_{\bar{w}} v(w) - F_0(w,\bar{w}) \\ &= -\frac{1}{4\pi}\bar{w}^{k-1}(k-\pi|w|^2)e^{-\pi|w|^2} - \left(-\frac{1}{4\pi}\bar{w}^{k-1}e^{-\pi|w|^2}\right) \\ &= \frac{1}{4\pi}\bar{w}^{k-1}e^{-\pi|w|^2}\left[-(k-\pi|w|^2)+1\right] \\ &= \frac{1}{4\pi}\bar{w}^{k-1}e^{-\pi|w|^2}(1-k+\pi|w|^2), \end{aligned}$$

which is (8). □

10 Constancy of q

Let $H(w) = \overline{q(w)}$. Define $Q(w) = H(w)/w^{k-1}$.

Lemma 10.1. *Assume*

$$q(w) = O(|w|^{k-1}) \qquad \text{as } w \to 0.$$

Then the holomorphic function $H(w) = \overline{q(w)}$ satisfies $H(w) = O(|w|^{k-1})$ as $w \to 0$. In particular, $Q(w) := H(w)/w^{k-1}$ extends to a holomorphic function on all of U.

Proof. The first assertion is immediate by complex conjugation. The holomorphic extension of Q across 0 then follows from the removable-singularity theorem. □

Proposition 10.2. *The function Q is constant in U.*

Proof. By Lemma 10.1, $Q = H/w^{k-1}$ is holomorphic in U. Therefore both $\Re Q$ and $\Im Q$ are harmonic in U.

On ∂U, (8) gives

$$H(w) = \overline{q(w)} = \frac{1}{4\pi}w^{k-1}e^{-\pi|w|^2}(1-k+\pi|w|^2),$$

so

$$Q(w) = \frac{H(w)}{w^{k-1}} = \frac{1}{4\pi}e^{-\pi|w|^2}(1-k+\pi|w|^2),$$

which is real-valued on ∂U. Hence $\Im Q = 0$ on ∂U.

Since $\Im Q$ is harmonic in U and vanishes on ∂U, the maximum principle gives $\Im Q \equiv 0$ in U. A real-valued holomorphic function on a connected open set is constant, so $Q \equiv C$ for some $C \in \mathbb{R}$. □

11 Geometric consequence for the boundary

Proposition 11.1. *∂U is contained in a union of at most two circles centered at the origin.*

Proof. Since $Q \equiv C$ in U, restricting to the boundary and using the formula for Q on ∂U from the proof of Proposition 10.2,

$$e^{-\pi|w|^2}(1-k+\pi|w|^2) = 4\pi C \qquad \text{for all } w \in \partial U. \tag{9}$$

This is a condition on $|w|$ alone. Consider the radial function

$$f(r) := e^{-\pi r^2}(1 - k + \pi r^2), \qquad r \geq 0.$$

Its derivative is

$$f'(r) = 2\pi r\, e^{-\pi r^2}(k - \pi r^2).$$

Thus f is strictly increasing on $(0, \sqrt{k/\pi})$ and strictly decreasing on $(\sqrt{k/\pi}, \infty)$, with $f(r) \to 0$ as $r \to \infty$. Therefore each value $4\pi C$ is attained at *at most two* radii $r > 0$.

Since (9) says every boundary point satisfies $f(|w|) = 4\pi C$, the boundary ∂U is contained in the union of at most two circles centered at the origin. □

12 Use of simple connectivity

Theorem 12.1. *Assume $U \subset \mathbb{C}$ is bounded, simply connected, and h_k is an eigenfunction. Assume moreover that the compactly supported solution u furnished by Proposition 2.2 satisfies*

$$u|_U \in C^{1,\alpha}(\overline{U})$$

for some $\alpha \in (0, 1)$, and that the anti-holomorphic function q from Definition 8.1 obeys

$$q(w) = O(|w|^{k-1}) \qquad \text{as } w \to 0.$$

Then U is a disk centered at the origin.

Proof. By Proposition 2.2, there exists a compactly supported distributional solution u of (3). By the standing assumption at the beginning of the note, one has $0 \in U$.

By Proposition 11.1, the boundary ∂U is contained in the union of at most two circles centered at the origin. Since U is simply connected, its boundary is connected. Therefore ∂U is contained in a *single* circle

$$|w| = R$$

for some $R > 0$.

Because ∂U is compact and connected, any proper connected subset of the circle $\{|w| = R\}$ would be a point or a closed arc, and such a set cannot be the boundary of a bounded domain. Hence in fact

$$\partial U = \{|w| = R\}.$$

Since $0 \in U$ and U is simply connected, the only bounded simply connected domain enclosed by this circle is the disk

$$U = \{|w| < R\}.$$

Therefore U is a disk centered at the origin. □

13 Summary of the mechanism

(1) Write the localization identity in Bargmann form (2).

(2) Use the Paley–Wiener divisibility lemma to obtain the compactly supported Poisson problem (3).

(3) Remove the pole $1/w^k$ via u_{sing}.

(4) Define $v = u - u_{\text{sing}}$, solving $\Delta v = \bar{w}^k e^{-\pi|w|^2}$.

(5) Identify the primitive F_0.

(6) Define the anti-holomorphic $q = \partial_{\bar{w}} v - F_0$.

(7) Assuming $q(w) = O(|w|^{k-1})$ at the origin, show that $Q(w) = \overline{q(w)}/w^{k-1}$ is holomorphic.

(8) Observe that Q is real on ∂U, hence constant.

(9) Conclude that ∂U is a circle.

14 Final remark

The argument extends the h_1 case at the level of the Poisson/free-boundary reduction and the boundary rigidity mechanism, but in the present note the final disk conclusion is conditional on two extra inputs: the boundary regularity needed to identify $\partial_{\bar{w}} u$ on ∂U, and the origin-vanishing condition $q(w) = O(|w|^{k-1})$ needed to divide by w^{k-1}.